COURSE SYLLABUS

ExInt II: Research Designs in SME Research (Group 3)

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1. Course Overview

This seminar is co-taught in two parts. Part A (Sessions 1–3) prepares students to write a research proposal. Part B (Sessions 4–5 and 7–8) focuses on implementing a fully reproducible empirical research project in a Python coding environment.

2. Course Outline

#	Date	Time	Room	Topic & Details
1	March 4, 2026	08:00 – 11:00	D1.1.078	**Logistics & Course Outline** Course logistics and administrative matters. Walkthrough of the syllabus. Overview of course structure, expectations, and assessment. Q&A.
2	March 18, 2026	08:00 – 11:00	D1.1.078	**Finding a Topic, Finding a Supervisor, Research Proposals & Thesis Guidelines** How to identify a research topic and find a suitable supervisor. Writing a research proposal: problem statement, research question, theoretical framework, methodology. ExInt master thesis guidelines and expectations.
3	April 15, 2026	08:00 – 11:00	D1.1.078	**Writing a Literature Review** Systematic literature search strategies and database use. Compiling and organising sources. Writing a coherent and critical literature review chapter.
4	May 6, 2026	08:00 – 11:00	D1.1.078	**Reproducible Research Setup & Research Question Development** Introduction to reproducible research principles. Python IDE setup (VS Code / JupyterLab). Project folder structure and version control basics. Reference management setup (Zotero integration, PDF storage). Students develop a research question within SME internationalization and derive a testable hypothesis from theory.
5	May 27, 2026	08:00 – 11:00	D1.1.078	**Data Acquisition via WRDS API & Exploratory Analysis** Connecting to WRDS via Python API. Querying firm-level panel data. Data cleaning, transformation, and variable construction. Exploratory data analysis and descriptive statistics. Students retrieve their datasets and begin preparing variables for regression analysis.
6	June 2, 2026	15:00 – 16:30	TC.2.01	**Dissertation Fair** Supervisors introduce themselves and present topics of potential interest. Students explore possible thesis directions and connect with prospective supervisors.

#	Date	Time	Room	Topic & Details
7	June 3, 2026	08:00 – 12:00	D1.1.078	**Time Series Regression Analysis & Results Documentation** Implementing time series / panel regression models in Python (statsmodels / linearmodels). Hypothesis testing and interpretation of results. Producing publication-quality tables and figures programmatically. Writing results directly in Markdown research note. In-class coaching and troubleshooting.
8	June 10, 2026	08:00 – 12:00	D1.1.078	**Research Note Finalization & Presentation** Completing the Markdown research note (introduction, theory, hypothesis, data, methodology, results, conclusion, references). Peer review of reproducibility (can a classmate re-run your project?). Final presentations: students walk through their coding environment and findings. Course synthesis and reflection.

Attendance Policy

Attendance is mandatory for all sessions. Students may miss a maximum of two sessions across the entire course (Parts A and B combined). Exceeding this limit will result in a failing grade, regardless of other performance.

3. Part A: Write a Research Proposal

Learning Outcomes

Upon successful completion of Part A, students will be able to:

- Identify a suitable research topic and approach a thesis supervisor whose expertise aligns with their interests.
- Write a structured research proposal following the ExInt guidelines, clearly articulating the research question, theoretical grounding, and intended methodology.
- Conduct a systematic literature review: searching databases, compiling sources, and writing a coherent review chapter.
- Use AI-assisted research tools (Google AI, NotebookLM, AntiGravity) to support the thesis preparation process.

Detailed Session Plan

Session 1 – Logistics & Course Outline (March 4)

Objectives: Introduce the course, explain the syllabus, and answer questions.

Topics covered:

- Course logistics and administrative matters.
- Walkthrough of the syllabus: course structure, sessions overview, and assessment criteria.
- Expectations for Part A and Part B, including deliverables and deadlines.
- Open Q&A session.

Session 2 – Finding a Topic, Finding a Supervisor, Research Proposals & Thesis Guidelines (March 18)

Objectives: Learn how to identify a research topic, find a supervisor, write a research proposal, and understand the ExInt master thesis guidelines.

Topics covered:

- Strategies for identifying a research topic in SME internationalization.
- Finding and approaching a thesis supervisor: matching research interests, preparing a brief pitch.
- Writing a research proposal: problem statement, research question, theoretical framework, and methodology outline.
- ExInt master thesis guidelines: structure, formatting, and quality expectations.

Deliverable by end of session: A preliminary topic description and a first draft of the research proposal (2–3 pages).

Session 3 – Writing a Literature Review (April 15)

Objectives: Learn how to search for, compile, and write a literature review.

Topics covered:

- Systematic literature search: using academic databases, defining search terms, and applying inclusion/exclusion criteria.
- Compiling and organising sources: reference management with Zotero, categorising literature, and identifying themes.
- Writing the literature review: structuring the narrative, synthesising findings, and positioning the research contribution.
- Using AI tools (Google AI, NotebookLM, AntiGravity) to support the literature review process.

Deliverable by end of session: A structured outline of the literature review with at least 10 relevant sources organised by theme.

Assessment

Students are required to write a research proposal (maximum 20 pages) that includes the following sections:

- Title page
- Table of Contents
- Introduction
- Literature Review
- Planned Methodology
- Expected Results
- References (minimum 30)
- Timetable

Formatting Guidelines

- Main body text: Arial, 11 pt, 1.5 line spacing
- Tables and figures: numbered and labelled, Arial 11 pt, single line spacing
- Citation style: APA (recommended)

Assessment Criteria

Section	Weight
Introduction	20 %
Literature Review	50 %
Planned Methodology	10 %
Expected Results	10 %
References	10 %

Submission Deadline

Submit your research proposal as a PDF file by **May 5th, 2026, 11:59 PM**.

Tools and Prerequisites (Part A)

Tool	Details
Gmail Account	Required for access to Google AI tools and collaboration
Google AI for Students	AI-powered research assistant for brainstorming and exploration
NotebookLM	AI notebook for synthesising and organising research sources
AntiGravity	AI-powered academic writing and research tool
Zotero	Reference manager for storing PDFs, organising sources, and generating citations
Silvi.ai	AI-powered academic writing assistant for structuring and drafting research proposals

4. Part B:

Learning Outcomes

Upon successful completion of Part B, students will be able to:

- Set up and manage a reproducible research project environment in Python, including folder structure, dependency management, and version control.
- Formulate a research question grounded in SME internationalization theory and derive a testable hypothesis that logically follows from theoretical arguments.
- Programmatically acquire firm-level panel data via the WRDS API and prepare it for empirical analysis.
- Implement and interpret time series regression models using Python (statsmodels / linearmodels).
- Manage academic references using a reference manager with locally stored PDFs, integrated into the research workflow.
- Produce a complete research note written directly in Markdown, including all standard sections of an empirical paper.
- Demonstrate full reproducibility: a peer should be able to re-run the entire project from data acquisition to final output.

Detailed Session Plan

Session 4 – Reproducible Research Setup & Research Question (May 6)

Objectives: Establish the complete working environment and formulate the research question and hypothesis.

Topics covered:

- Principles of reproducible research: why it matters, what it looks like in practice.
- IDE setup together in class: Python environment (VS Code or JupyterLab), virtual environment, required packages (pandas, statsmodels, linearmodels, wrds, matplotlib, etc.).
- Project folder structure: /data, /code, /references, /output, README.md.
- Reference management: setting up Zotero (or similar), storing PDFs in /references, citing in Markdown via BibTeX.
- Research question development: identifying a gap in SME internationalization literature, grounding the question in theory.
- Hypothesis derivation: translating theoretical arguments into a testable, falsifiable statement.

Deliverable by end of session: Working Python environment, initialized project folder, and a draft research question with at least one hypothesis documented in README.md.

Session 5 – Data Acquisition via WRDS API & Exploratory Analysis (May 27)

Objectives: Connect to WRDS, retrieve firm-level panel data, and conduct exploratory analysis.

Topics covered:

- Introduction to WRDS: available databases, firm identifiers, navigating the data catalog.
- Python WRDS library: authentication, connection, SQL queries to extract firm-level data.
- Selecting appropriate variables for time series regression (dependent, independent, control variables).
- Data cleaning: handling missing values, outliers, merging datasets, constructing panel structure.
- Exploratory data analysis: descriptive statistics, correlation matrices, time series plots.
- Saving cleaned data to /data with documentation of all transformations.

Deliverable by end of session: A reproducible data acquisition script, cleaned dataset, and exploratory analysis notebook with descriptive statistics.

Session 7 – Time Series Regression Analysis & Results (June 3)

Objectives: Estimate regression models, test hypotheses, and begin documenting results in the research note.

Topics covered:

- Panel data regression with Python: OLS, fixed effects, random effects (linearmodels.PanelOLS).

- Model specification: choosing between FE/RE (Hausman test), clustering standard errors, robustness checks.
- Hypothesis testing: interpreting coefficients, p-values, confidence intervals in relation to the derived hypothesis.
- Producing publication-quality output: regression tables (stargazer-style via Python), figures (matplotlib/seaborn).
- Writing results directly in Markdown: structuring the methodology and results sections of the research note.
- Individual coaching and troubleshooting during working time.

Deliverable by end of session: Estimated regression models with results, formatted tables and figures, and a draft methodology + results section in the Markdown research note.

Session 8 – Research Note Finalization & Presentation (June 10)

Objectives: Complete the research note, verify reproducibility, and present findings.

Topics covered:

- Completing the Markdown research note: introduction, theoretical framework, hypothesis, data description, methodology, results, discussion, conclusion, and references.
- Integrating BibTeX references from the reference manager into the final note.
- Reproducibility check: peer exchange – can a classmate clone your project and reproduce the results end-to-end?
- Final presentations: students walk through their coding environment, demonstrate the pipeline from data acquisition to output, and discuss their findings.
- Course synthesis: reflection on the reproducible research workflow and its value for future thesis work.

Final deliverable: A complete, reproducible research project consisting of: (1) all code in the project folder, (2) the Markdown research note as the primary output file, (3) references as PDFs with BibTeX, and (4) a successful reproducibility demonstration.

Required Project Structure

Each student's project must follow this standardized folder structure to ensure reproducibility:

Path	Purpose
`project-root/`	Top-level project directory
`├── README.md`	Project description, research question, hypothesis, setup instructions
`├── requirements.txt`	Python dependencies for reproducibility
`├── code/`	All Python scripts and notebooks
`│   ├── 01_data_acquisition.py`	WRDS API query and raw data download
`│   ├── 02_data_cleaning.py`	Cleaning, merging, variable construction
`│   ├── 03_analysis.py`	Regression models and hypothesis tests
`│   └── 04_figures_tables.py`	Publication-quality output generation
`├── data/`	Raw and processed datasets (not shared if proprietary)

`├── references/`	PDF articles + references.bib (BibTeX file)
`├── output/`	Generated tables, figures, and final research note
`└── output/research_note.md`	The final Markdown research note

Research Note Structure (Markdown)

The research note (output/research_note.md) must contain the following sections:

1. Title and author information
2. Abstract (150–250 words)
3. Introduction: motivation, research gap, and research question
4. Theoretical framework and literature review: key theories and prior findings
5. Hypothesis development: clearly stated hypothesis with explicit derivation from theory
6. Data and sample description: source (WRDS), sample period, firm selection criteria, variable definitions
7. Methodology: regression specification, estimation approach, robustness strategy
8. Results: regression output, hypothesis test outcomes, interpretation
9. Discussion and conclusion: implications for research and practice, limitations, future research directions
10. References: generated from references.bib via citation keys in Markdown

Assessment

Part B (Sessions 4–5 & 7–8):

Component	Weight	Description
Reproducible Research Project	40%	Complete project folder with code, data pipeline, and reproducible results. Assessed on code quality, proper structure, and end-to-end reproducibility.
Markdown Research Note	30%	Written research note covering all required sections. Assessed on clarity of research question, quality of hypothesis derivation from theory, rigor of analysis, and proper referencing.
Final Presentation	30%	Live walkthrough of the coding environment and findings. Assessed on ability to explain and defend methodological and analytical choices.

Tools and Prerequisites (Part B)

Tool	Details
Python 3.10+	Core programming language for all project work
VS Code / JupyterLab	IDE for development (set up together in Session 5)

WRDS Account	Students must have an active WRDS account (institutional access via WU)
Key Python Packages	pandas, numpy, statsmodels, linearmodels, wrds, matplotlib, seaborn, stargazer
Zotero (or similar)	Reference manager for storing PDFs and generating BibTeX
Git (recommended)	Version control for tracking changes and enabling reproducibility